

# Lionael Surya Dwitama

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## PROFESSIONAL SUMMARY

Data Science and Analytics student at Seneca Polytechnic (3.9 GPA, President's Honour List) who builds working systems, not just models. Experience spans the full pipeline — data ingestion, feature engineering, model training, and deployment — using Python, SQL, and XGBoost, with model explainability (SHAP) and rigorous validation (walk-forward testing, Spearman checks) built in rather than added after the fact. Has shipped models behind FastAPI/Streamlit interfaces and combined ML with LLM-based explanation layers on real-world data.

## EDUCATION

### Seneca Polytechnic — Honours Bachelor of Data Science and Analytics

Toronto, ON

President's Honour List (Winter 2025) | Cumulative GPA: 3.9/4.0

Expected Apr. 2028

- Relevant coursework: Machine Learning, Statistical Methods, Data Mining, Database Systems, High Performance Computing, Data Visualization, Business Intelligence

## TECHNICAL SKILLS

**Languages:** Python, SQL, JavaScript

**ML & Statistics:** XGBoost, SHAP, Prophet, scikit-learn, walk-forward validation, PCA, hypothesis testing

**Data Engineering:** Apache Kafka, Apache Spark, Delta Lake, DuckDB, dbt, PostgreSQL, MySQL, Supabase

**Deployment & Tools:** FastAPI, Streamlit, Docker, Git/GitHub, Power BI, pytest

## PROJECTS

### Customer Churn Prediction — Calibrated XGBoost with ROI-Optimized Thresholds

- Built an end-to-end churn model using calibrated XGBoost, selecting the decision threshold to maximize retention-offer ROI rather than default accuracy
- Applied SHAP to explain individual predictions and surface the top churn drivers for the business team
- Deployed the model behind a FastAPI backend with a Streamlit interface, and added drift monitoring to flag when the model needed retraining

### IDX Stock Signal Scanner — Rules → ML → LLM Explanation Pipeline

- Designed a three-layer equity screening pipeline: a rules-based filter to narrow candidates, an XGBoost model to rank them, and a Claude API layer to generate plain-language explanations for each signal
- Built the underlying data pipeline to pull and process Indonesian equity market data for the ranking model

### Retail Demand Forecasting — Model Benchmarking with Walk-Forward Validation

- Benchmarked naive seasonal, Prophet, and XGBoost forecasting approaches using walk-forward validation to simulate real deployment conditions
- Evaluated trade-offs in accuracy, interpretability, and compute cost across the three approaches to recommend a forecasting strategy

### Nutrition Priority Index — Indonesia

- Compared a PCA-based composite index against single-indicator methodology on real Indonesian government nutrition data, validating results with Spearman correlation
- Enforced data quality with 67 pytest tests and YAML data contracts, and shipped findings through an interactive Streamlit dashboard

## EXPERIENCE

### QuickRN Canada

Toronto, ON

Web Development and Data Operations

Jul. 2025 – Present

- Build and ship features for Aspen OS, an internal operations platform (Next.js, TypeScript, Supabase) that runs the company's day-to-day workflows in production
- Diagnose and resolve website and online course access issues end-to-end, cutting the back-and-forth between users and the dev team by handling root-cause investigation directly

### CV. Veda Sakti Dharma

Jakarta, Indonesia

Software Engineering Intern

Jun. 2025 – Aug. 2025

- Automated PPh 21 payroll tax calculations in Python, removing a manual, error-prone step from the payroll workflow
- Validated every automated output against source payroll records before rollout, catching discrepancies before they reached final reporting

## LEADERSHIP & ACTIVITIES

### PERMIKA Toronto — IT and Media Strategy Associate

- Own social media campaign strategy and execution for the Indonesian student organization at Seneca Polytechnic, from content planning through publishing